

Workshop Overview

This hands-on workshop teaches you how to design, develop, and deploy enterprise-grade applications using agentic coding with Kiro. Traditionally, building and deploying sophisticated applications requires dedicated development and DevOps resources. Through this workshop, you'll discover how agentic coding transforms this process, enabling you to create production-ready applications through natural conversation and collaboration with AI.

What You'll Learn

By the end of this workshop, you will have:

- **Built a complete AI-powered chatbot** with text, image, and voice capabilities
- **Deployed secure, scalable infrastructure** on AWS using serverless architecture
- **Implemented enterprise security practices** including user authentication and authorization
- **Mastered agentic coding techniques** for rapid application development
- **Gained hands-on experience** with modern cloud services and AI integration
- **Created a reusable solution package** using infrastructure as code

Learning Objectives

Technical Skills

- Design and deploy serverless applications on AWS
- Integrate AI capabilities using Amazon Bedrock foundation models
- Implement secure user authentication with Amazon Cognito
- Apply security best practices and enterprise patterns
- Package solutions using AWS CDK for reusability

Agentic Coding Mastery

- Collaborate effectively with AI development tools
- Provide clear requirements and iterate on feedback
- Navigate the development process from concept to production
- Troubleshoot and refine applications through AI assistance

Target Audience

This workshop is designed for:

- **Product managers** who want to rapidly prototype and validate ideas
- **Entrepreneurs** looking to build MVPs without extensive technical teams
- **Technical professionals** interested in AI-assisted development
- **Anyone with a passion** for building software products and services

Prerequisites

Required

- **AWS Account** with appropriate permissions
- **Kiro IDE** installed and configured
- **Basic computer literacy** and comfort with web interfaces

Helpful but Not Required

- Previous coding experience (nice-to-have)
- Familiarity with AWS services (will be explained)
- Understanding of web applications (will be covered)

Workshop Structure

The workshop is divided into two main sections:

Vibe Coding (Exercises 1-10)

Learn intuitive, conversational development through hands-on exercises that progressively build a complete AI chatbot application.

Spec Coding (Advanced)

Explore structured, specification-driven development for complex enterprise applications.

Time Commitment

- **Total Duration:** 6-10 hours
- **Focused Time:** 5-6 hours of active development
- **Flexible Pacing:** Complete at your own speed with natural break points
- **Modular Design:** Each exercise builds on the previous, with clear checkpoints

What You'll Build

Throughout this workshop, you'll create a sophisticated AI chatbot featuring:

- **Secure Frontend:** Hosted on AWS with global CDN
- **Serverless Backend:** Auto-scaling Lambda functions and API Gateway
- **AI Intelligence:** Text and image generation using Amazon Bedrock
- **User Authentication:** Enterprise-grade security with Amazon Cognito
- **Multi-modal Capabilities:** Text, voice, and visual interactions
- **Production-Ready Package:** Complete CDK solution for deployment

Getting Started

Ready to transform your ideas into reality? Let's begin your journey into agentic coding and discover how quickly you can build enterprise-grade applications with Kiro as your development partner.